

360-Degree Monitoring Platform - Business Requirements

1. Business need

Operations teams require a single, repeatable way to detect service degradation, understand probable cause and act before or during an outage. The solution must correlate host health, endpoint availability, application telemetry and logs sufficiently for a learner/operator to diagnose a controlled failure.

2. Goals

- reduce blind spots in host/application health;
- detect actionable failures quickly;
- centralize evidence used for diagnosis;
- make dashboards/alerts reproducible as code/configuration;
- provide documented recovery actions;
- demonstrate open-source observability principles.

3. Scope

In scope

- Linux host metrics;
- application or web endpoint availability;
- centralized logs;
- dashboards;
- alert rules and notification routing;
- deployment automation;
- runbooks;
- controlled failure/recovery test.

Out of scope by default

- enterprise multi-region HA;
- petabyte-scale log retention;
- paid SaaS monitoring dependency;
- production PII/customer data.

4. Functional requirements

ID	Requirement	Priority
MON-FR-001	Collect CPU, memory, filesystem and basic network/system metrics from at least one Linux host.	Must
MON-FR-002	Monitor at least one service/HTTP endpoint using an availability probe.	Must
MON-FR-003	Centralize logs from the monitored workload or host.	Must
MON-FR-004	Provide at least one host dashboard and one service dashboard.	Must
MON-FR-005	Create alerts for at least three operational symptoms, including service unavailable and resource pressure.	Must
MON-FR-006	Route alerts to a configured notification target suitable for the training environment.	Must
MON-FR-007	Attach a runbook reference or operator instruction to critical alerts.	Must

ID	Requirement	Priority
MON-FR-008	Preserve dashboards and alert rules in version-controlled configuration where the chosen tooling supports it.	Must
MON-FR-009	Expose health status for the monitoring components themselves.	Must
MON-FR-010	Demonstrate a controlled failure and recovery while telemetry captures the incident.	Must
MON-FR-011	Record deployment/release annotations or equivalent change context on dashboards.	Should
MON-FR-012	Collect application metrics and/or traces through OpenTelemetry.	Should
MON-FR-013	Provide service-level indicators such as availability, latency or error rate.	Should
MON-FR-014	Support more than one monitored host through repeatable configuration.	Should

5. Non-functional requirements

ID	Area	Requirement
MON-NFR-001	Security	Monitoring endpoints and dashboards must not expose credentials; administrative access must be protected.
MON-NFR-002	Repeatability	Deployment must be reproducible using documented automation such as Docker Compose, Ansible, Terraform or Kubernetes manifests.
MON-NFR-003	Reliability	Restarting monitoring components must not require manual recreation of dashboards/rules.
MON-NFR-004	Operability	A new operator must be able to identify the source of a seeded incident using the documented dashboards/logs/runbook.
MON-NFR-005	Alert quality	Alerts must be actionable and avoid duplicate/noisy rules for the same symptom.
MON-NFR-006	Performance	Monitoring overhead must not materially destabilize the monitored training workload.
MON-NFR-007	Retention	Metric/log retention must be documented and appropriate for the available lab storage.
MON-NFR-008	Portability	The reference implementation should run on a standard Linux VM or equivalent lab environment without proprietary agents.

6. Required incident scenarios

At least two scenarios must be tested:

1. **Service outage:** stop or break the monitored service; verify probe failure, alert, logs/evidence and recovery.
2. **Resource symptom:** create controlled CPU, memory or disk pressure; verify dashboard visibility and alert behavior.

Optional scenarios: - certificate expiry warning; - elevated application errors; - slow endpoint/latency; - disk-filling condition; - failed deployment.

7. Operational acceptance

The project passes only if the team can diagnose the seeded incident using the monitoring platform without first reading the failure-injection command.